

PROMPT-SWITCHED CO-GENERATION FOR WORLD ACTION MODELS

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ABSTRACT

World action models predict what a robot will see next in addition to what it will do, but “what it will see” is almost always RGB, while the quantities robot software consumes — metric depth, dense semantics, surface orientation — are delegated to separate stacks with their own heads and losses. We ask whether one video-diffusion policy can emit all of them, selected by words, with no extra parameter, head or loss term. Our recipe, COGEN, has three parts: every signal, the action included, is rendered into the RGB space the frozen video VAE already models, through invertible codecs whose fidelity is verified before training; a *task template* names the modality set occupying the packed latent sequence and is announced verbatim in the prompt; and a *conditioning mask*, drawn independently of the template, decides which segments are given, which makes “perception” and “world model” two draws from one distribution rather than two systems. On 16 RL Bench tasks, one 5B checkpoint fine-tuned for 4.5k steps serves four streams from four prompts: depth at 0.059–0.160 AbsRel, nine-role dense segmentation at 0.33–0.59 macro-mIoU, normals at 0.90–0.94 mean cosine, and end-effector trajectories at 1.9–5.4 cm median error, while remaining a policy. We further show that the conditioning mixture, not just the task mixture, decides which questions a finished checkpoint can still answer, and that diffusion loss does not order checkpoints by closed-loop success: the checkpoint whose success rate fell $5\times$ had the lowest training loss of the entire run.

1 INTRODUCTION

A world action model (WAM) couples action prediction with future-observation prediction, on the premise that a policy that can imagine the consequences of acting is a better policy (Zhu et al., 2025; Zhen et al., 2026). In practice the imagined future is RGB video — a strange restriction, since downstream robot software rarely consumes RGB directly. It consumes depth for collision checking and grasp scoring, role masks for tracking the objects a task is about, and surface orientation for approach reasoning. Recent WAMs acknowledge this by attaching modality-specific decoders (Ma et al., 2026; Yuan et al., 2026a; Zhao et al., 2026), which fixes the modality set at architecture time and adds a loss weight and a capacity budget per head.

A separate literature argues the opposite structural point: image and video generators, prompted appropriately, are already generalist vision learners (Gabeur et al., 2026; Wang et al., 2026). If a model can be *asked in text* for a depth map and returns a decodable one, no depth head was necessary. The same move appears in driving (Yuan et al., 2026b) and, in pure vision, as a compact multitask model whose task set is fixed at pretraining (Zhuang et al., 2025).

Action Images (Zhen et al., 2026) supplies the bridge: an *action* can be drawn, by projecting a 7-DoF end-effector pose through each camera into a three-channel Gaussian canvas, so action generation becomes video generation and needs no action head. Once actions are images, the type distinction between “what the policy outputs” and “what a perception model outputs” disappears. Everything is a video segment, and only two questions remain: which segments are present, and which are given. We take that literally and build COGEN, in which action, depth, dense segmentation, normal and RGB are five interchangeable occupants of one packed latent sequence, denoised by one shared head under one loss. Our contributions:

- **Two orthogonal control axes.** We factor a sample into a template Π (which modalities occupy the sequence, announced in the prompt) and a condition set $\mathbb{G}$ (which latent positions are clean). This makes explicit something the literature conflates: *perception is not a modality, it is a $(\Pi, \mathbb{G})$ pair*. The template {video, depth} with RGB given is monocular depth estimation; the same template with only first frames given is joint RGB–depth world modelling; and {depth, action} contains no RGB at all, so it is a depth-space world model with no perception content (section 4.3).
- **An admission gate for new modalities.** A modality is trainable here iff it can be written as an image the frozen VAE preserves and read back as a metric quantity. We make that a measured precondition rather than an assumption (section 4.1).
- **Auditable mixtures.** Prompt-switched multitask training fails *silently*: a missing annotation turns a sample into a different task while the loss stays smooth. We give the invariants that convert such drift into startup errors, and quantify one we hit (section 4.5).
- **Results.** One checkpoint serves four modalities on 16 RL Bench tasks; the conditioning mixture is shown to decide which modes survive training; and diffusion loss is shown not to order checkpoints by closed-loop success (section 5).

2 RELATED WORK

WAMs with specialised heads. GeoSem-WAM (Ma et al., 2026) predicts geometry- and semantics-aware future states alongside actions, DreamWAM (Yuan et al., 2026a) argues explicitly for going beyond RGB futures, and RynnWorld-4D (Zhao et al., 2026) predicts 4D futures for manipulation. All attach modality-specific decoders to a shared trunk, fixing the modality set at architecture time. COGEN spends zero new parameters and selects the modality from the prompt, so the task set is a property of the data mixture rather than of the network.

Unifying world states and actions. UWM (Zhu et al., 2025) couples video and action diffusion with per-stream timesteps. We share the motivation but not the mechanism: rather than two diffusion processes over two representations, there is one process over one representation, because the action is already an image (Zhen et al., 2026). Adding a fifth stream therefore costs one codec, not one branch.

Understanding as generation. Image (Gabeur et al., 2026) and video (Wang et al., 2026) generators have been shown to perform depth, segmentation and correspondence from text prompts alone; SUV (Yuan et al., 2026b) makes the same move for driving. Argus (Zhuang et al., 2025) pursues compact multitask vision with per-task adapters and reports — a finding we treat as a design constraint — accuracy loss on *every* task once its core task set outgrows its data.

The gap. The generation-as-understanding literature has no actions; the WAM literature has per-modality heads. The intersection — one head, one loss, prompt-switched, and *including the action stream* — raises a question neither parent asks: under which conditioning pattern should an auxiliary stream be trained, when deployment only ever uses one of them.

3 PRELIMINARIES

This section fixes the backbone, the action-image encoding, and the objective, introducing each symbol where it first appears. Two conventions are worth flagging because they are where a reader of the diffusion literature would otherwise guess wrong: n indexes *video frames* while t indexes *flow-matching time*, and a single t is drawn per training step and shared by every segment of the packed sequence — unlike UWM (Zhu et al., 2025), which gives each stream its own timestep.

3.1 BACKBONE AND LATENT SPACE

We build on a text-conditioned latent video diffusion transformer (Wan2.2-TI2V-5B, Wan Team, 2025) with a frozen causal 3D VAE, whose encoder and decoder we write $\mathcal{E}$ and $\mathcal{D}$. A clip of T

frames at $H \times W$ pixels encodes to a grid of latents,

$$\mathcal{E} : \mathbb{R}^{3 \times T \times H \times W} \longrightarrow \mathbb{R}^{C_l \times T_l \times H_l \times W_l}, \quad T_l = 1 + \frac{T-1}{4}, \quad H_l = \frac{H}{16}, \quad W_l = \frac{W}{16}, \quad (1)$$

where C_l is the number of latent channels and (T_l, H_l, W_l) the latent resolution; the constants 4 and 16 are the VAE’s temporal and spatial compression ratios. In the configuration of section 5.1, $T = 41$ and $H = W = 512$ give $C_l = 48$ and an $11 \times 32 \times 32$ latent grid. Only the transformer, written f_θ with parameters θ , is trained; $\mathcal{E}$ and $\mathcal{D}$ stay frozen throughout, which is what turns the codecs of section 4.1 into a *data* question rather than a training question.

Each sample packs V camera views, indexed $v \in \{0, \dots, V-1\}$; we use $V = 2$. View v has intrinsics $\mathbf{K}_v \in \mathbb{R}^{3 \times 3}$ and camera-to-world extrinsics $\mathbf{E}_v \in \mathbb{R}^{4 \times 4}$, which together induce the projection $\pi_v : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ from a world point to a pixel coordinate. Multi-view conditioning follows Zhen et al. (2026), in the lineage of camera-controlled video generation (Bai et al., 2025): for each view we build the six-channel Plücker ray map (Sitzmann et al., 2021) of every pixel, split it into its direction and moment halves, encode each half with $\mathcal{E}$, and concatenate them along the channel axis. Stacking the result over segments gives the camera conditioning tensor $\mathbf{C}$, injected into every transformer block through a zero-initialised encoder, so an untrained camera pathway is exactly the identity. One detail is load-bearing and easy to get wrong: the poses generating $\mathbf{C}$ are *relative* to view 0 at frame 1, whereas the extrinsics $\mathbf{E}_v$ behind π_v are *absolute* camera-to-world. They are different quantities and are not interchangeable.

3.2 ACTIONS AS IMAGES

An action image renders an end-effector trajectory into the same pixel space as video, so that generating an action and generating a frame become the same operation (Zhen et al., 2026). Let $n \in \{1, \dots, T\}$ index frames, and let the gripper state at frame n be its position $\mathbf{p}_n \in \mathbb{R}^3$, orientation $\mathbf{R}_n \in SO(3)$ and binary openness $g_n \in \{0, 1\}$. Three image points are formed — the gripper centre, and the tips of two axes carried a distance $\ell = 0.1$ m along the gripper’s current orientation:

$$\mathbf{q}_n^{(1)} = \pi_v(\mathbf{p}_n), \quad \mathbf{q}_n^{(c)} = \pi_v(\mathbf{p}_n + \ell \mathbf{R}_n \mathbf{e}_c), \quad c \in \{2, 3\}, \quad (2)$$

where $\mathbf{e}_2 = (1, 0, 0)^\top$ and $\mathbf{e}_3 = (0, 0, -1)^\top$ are fixed unit axes of the gripper frame. Each point is painted into its own colour channel as an isotropic Gaussian, giving the action image $\mathbf{A}^{(v)} \in [0, 1]^{3 \times T \times H \times W}$ of view v with entries

$$A_{c,n,\mathbf{x}}^{(v)} = \exp\left(-\frac{\|\mathbf{x} - \mathbf{q}_n^{(c)}\|_2^2}{2\sigma^2}\right), \quad \sigma = 0.05 \min(H, W), \quad (3)$$

in which $\mathbf{x}$ ranges over pixel coordinates and σ is the blob width in pixels. Channels $c = 1, 2, 3$ are R, G and B, so red carries position and green and blue carry the two orientation axes. The binary gripper state has no natural spatial extent, so it is written as a pedestal in the blue channel wherever that channel’s own blob is weak:

$$A_{3,n,\mathbf{x}}^{(v)} \leftarrow \begin{cases} A_{3,n,\mathbf{x}}^{(v)}, & \text{if } A_{3,n,\mathbf{x}}^{(v)} > \frac{1}{4}, \\ \frac{1}{4} \mathbf{1}_{g_n > 1/2}, & \text{otherwise,} \end{cases} \quad (4)$$

with $\mathbf{1}$ the indicator function; this is the encoding of Zhen et al. (2026, Eq. 7). The image is finally rescaled to the VAE’s input range and encoded, $\mathbf{Z}^{(\text{action},v)} = \mathcal{E}(2\mathbf{A}^{(v)} - 1)$, where $\mathbf{Z}^{(m,v)}$ denotes the latent segment of modality m in view v .

Decoding inverts equation 3: the three blobs are located per view, and the 3D pose is recovered by sweeping candidate depths along the camera ray of $\mathbf{q}^{(1)}$ and keeping the depth at which the views agree best. Two properties of this construction matter later. The canvas is rendered *inside* the forward pass from the trajectory and both cameras, so one 3D trajectory yields two view-consistent action segments at no annotation cost; and because the result is an ordinary RGB video, an action segment is indistinguishable from a video segment as far as f_θ is concerned.

3.3 THE PACKED SEQUENCE AND ITS OBJECTIVE

The recipe of Zhen et al. (2026) concatenates four segments along the latent-time axis, $[V_0 \mid A_0 \mid V_1 \mid A_1]$, writing V_v and A_v for the RGB and action segments $\mathbf{Z}^{(\text{video},v)}$ and $\mathbf{Z}^{(\text{action},v)}$ of view

v . We write $\mathbf{Z}$ for the resulting packed sequence, L for its length along the latent-time axis, and $\mathbf{Z}_i$ for its i -th latent frame, i.e. the slice $\mathbf{Z}_{:,i,:} \in \mathbb{R}^{C_l \times H_l \times W_l}$. Positions split into two sets: the *given* positions $\mathbb{G} \subseteq \{1, \dots, L\}$, whose latents are supplied clean as conditions, and the *predicted* positions $\mathbb{P} = \{1, \dots, L\} \setminus \mathbb{G}$, which the model must produce.

Training draws a flow-matching time $t \in (0, 1]$ and Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ of the same shape as $\mathbf{Z}$, forms the linear interpolant between data and noise, and then restores the given positions to their clean values:

$$\tilde{\mathbf{Z}}_t = (1 - t)\mathbf{Z} + t\epsilon, \quad [\tilde{\mathbf{Z}}_t]_i = \mathbf{Z}_i \text{ for } i \in \mathbb{G}. \quad (5)$$

The regression target is the constant straight-line velocity of that interpolant, $\mathbf{U} = \epsilon - \mathbf{Z}$, and the loss is taken over the predicted positions only:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, \epsilon} \left[w(t) \frac{1}{|\mathbb{P}| C_l H_l W_l} \sum_{i \in \mathbb{P}} \left\| [f_\theta(\tilde{\mathbf{Z}}_t, t, \mathbf{C}, c)]_i - \mathbf{U}_i \right\|_2^2 \right], \quad (6)$$

where c is the prompt embedding, $\mathbf{C}$ the camera conditioning of section 3.1, and $w(t)$ a per-timestep weight. Concretely, t is drawn from the 1000-point shifted schedule $t = s\hat{t}/(1 + (s-1)\hat{t})$ with shift $s = 3$ and $\hat{t}$ uniform on $(0, 1]$, and $w(t)$ is that schedule’s bell-shaped weight, largest at mid-noise and normalised over the schedule. Excluding $\mathbb{G}$ from the sum, rather than merely feeding it clean, is what makes a condition segment cost no gradient: the model is never asked to reconstruct something it was given.

Everything in section 4 changes only *what the segments contain* and *which positions are in* $\mathbb{G}$. Equation 6 is untouched — there is no second loss term, no per-modality weight, and no modality-specific head.

4 METHOD

4.1 MODALITIES AS IMAGES, AND THE ADMISSION GATE

A modality is admissible iff (i) it encodes to a $[-1, 1]$ RGB video that the *frozen* VAE round-trips with low error and (ii) that video decodes back to the original metric quantity. Condition (ii) separates this from false-colour visualisation: the output must be a measurement, not a picture of one. We instantiate four streams; full codec definitions are in appendix A.

Depth. Metric depth d is compressed by the Barron power transform (Barron, 2019) and the result indexes a Gray-code Hamilton path over the eight RGB-cube vertices $\mathbf{v}_0, \dots, \mathbf{v}_7$ (Gabeur et al., 2026):

$$y = 1 - \left(1 - \frac{d}{\lambda c}\right)^{\lambda+1}, \quad \gamma(y) = (1 - \alpha)\mathbf{v}_k + \alpha\mathbf{v}_{k+1}, \quad k = \lfloor 7y \rfloor, \alpha = 7y - k, \quad (7)$$

with $\lambda = -3$, $c = 10/3$. Decoding projects a pixel onto the nearest of the seven edges and inverts equation 7; a pixel farther than a threshold from every edge decodes to *invalid* rather than to a plausible-looking depth. Spreading y over a length-7 curve rather than one channel is what makes this robust: VAE noise of a given magnitude costs $7\times$ less in y .

Surface normal and segmentation. Normals are derived from the same depth — no new data collection — by differentiating it with the intrinsics folded in, and stored as $\frac{1}{2}(\mathbf{n} + 1)$ (appendix A). Segmentation uses a *scene-role* protocol rather than a referring one: colour binds a cross-task functional role (target, goal, robot_arm, gripper, distractor, tool, fixture, background) identically in every task, episode, frame and view, so the palette is learnable from data and the prompt tag can be the constant `<scene-seg>`, carrying zero bits. The motivation over referring masks is density: a referring target occupies a median 0.18% of pixels, a role map 11.8%, and 0.18% is inside the VAE’s noise floor for much of an episode.

Table 1 reports the gate. It is cheap and decisive: a stream the VAE destroys cannot be rescued by training, so running the gate first converts a multi-day training question into a ten-minute data question.

Table 1: The admission gate. Each stream must survive a codec round-trip (uint8 quantisation only) and a round-trip through the frozen VAE *before* a run is launched; both are measured against the decoded ground-truth stream, so quantisation cancels in the VAE column. The RGB control is the same VAE on the same clips and bounds what any stream can reach. Depth and normal are 12 episodes spanning 12 tasks; the codec column for depth and normal is 6 episodes. *off-path* is the fraction of valid GT pixels the VAE pushes off the RGB-cube path, where decoding returns no depth at all rather than a wrong one — a cliff that an AbsRel over surviving pixels would hide.

Stream	Encoding	Codec round-trip	VAE round-trip
Depth	Barron $\rightarrow$ RGB-cube path, equation 7	AbsRel 0.066%	AbsRel 0.362% , 0.02% off-path
Normal	$\frac{1}{2}(\mathbf{n} + 1)$, equation 13	mean cos 0.999998	mean cos 0.9906 (7.9°)
Segm.	role LUT, 9-colour palette	IoU > 0.98 (spec)	target IoU median 0.965
Action	3 Gaussian channels, equation 3	≈ 4 mm 3D	not measured [†]
RGB control (same VAE, same clips)		n/a	29.6–31.5 dB PSNR

[†] the action stream is rendered in-loop rather than loaded, so its gate needs a different harness; see section 6.

4.2 TASK TEMPLATES: WHAT OCCUPIES THE SEQUENCE

A *template* is a subset Π of the five modalities, which carry a fixed total order

$$\text{video} \prec \text{depth} \prec \text{segmentation} \prec \text{normal} \prec \text{action}. \quad (8)$$

Writing $\mathbf{z}^{(m,v)}$ for the latents of modality m in view v and $\oplus$ for concatenation along the latent time axis, the sequence is laid out view-major and modality-minor,

$$\mathbf{z} = \bigoplus_{v=0}^{V-1} \bigoplus_{\substack{m \in \Pi \\ \text{in } \prec \text{ order}}} \mathbf{z}^{(m,v)}, \quad (9)$$

so video+depth over $V=2$ views packs as $[V_0 \mid D_0 \mid V_1 \mid D_1]$. Fixing the order equation 8 means <depth><action> and <action><depth> assemble identically, so no capacity is spent learning permutation invariance. The prompt is prefixed with the atomic tags of Π in the same order, e.g. "<video><depth> slide the bottom drawer open".

Two consequences make this cheap. *No new parameters*: a depth segment is a video segment as far as the transformer is concerned, so there is no depth head, no depth loss term and no loss weight to tune. *No new cost* for two-modality templates: an auxiliary modality takes the *action* slot rather than being appended, so video+depth has exactly the sequence length, memory footprint and step time of the official video+action. Equation 9 also admits three-modality templates such as video+depth+action, which generate perception and action in one sequence at $1.5\times$ the sequence length; that cell is expressible but is *not* trained in this work. Every arm below mixes two-modality templates only, so co-generation here means RGB together with one auxiliary stream, plus action in its own templates.

4.3 CONDITIONING MASKS: WHAT IS GIVEN

The second axis is the condition set $\mathbb{G}$. A *plan* assigns every segment (m,v) one flag $\phi_{m,v} \in \{\mathbb{F}, \mathbb{I}, 1\}$, from which $\mathbb{G}$ is built: $\mathbb{F}$ puts the whole segment into $\mathbb{G}$ (a pure condition — nothing is predicted from it), $\mathbb{I}$ puts only its first latent frame into $\mathbb{G}$, and 1 does the same after truncating the segment to a single latent frame. Assembly enforces

$$\mathbb{P} \neq \emptyset, \quad (10)$$

because a fully-conditioned sequence makes $|\mathbb{P}| = 0$ in equation 6; the implementation clamps that normaliser to 1, so the step reports a loss of exactly 0 — a full forward and backward that teaches nothing and looks like excellent convergence. This is also why $1\mathbb{I}1\mathbb{I}$ below collapses the RGB anchor only: in a perception template every segment is visual, so collapsing all of them would leave nothing to predict.

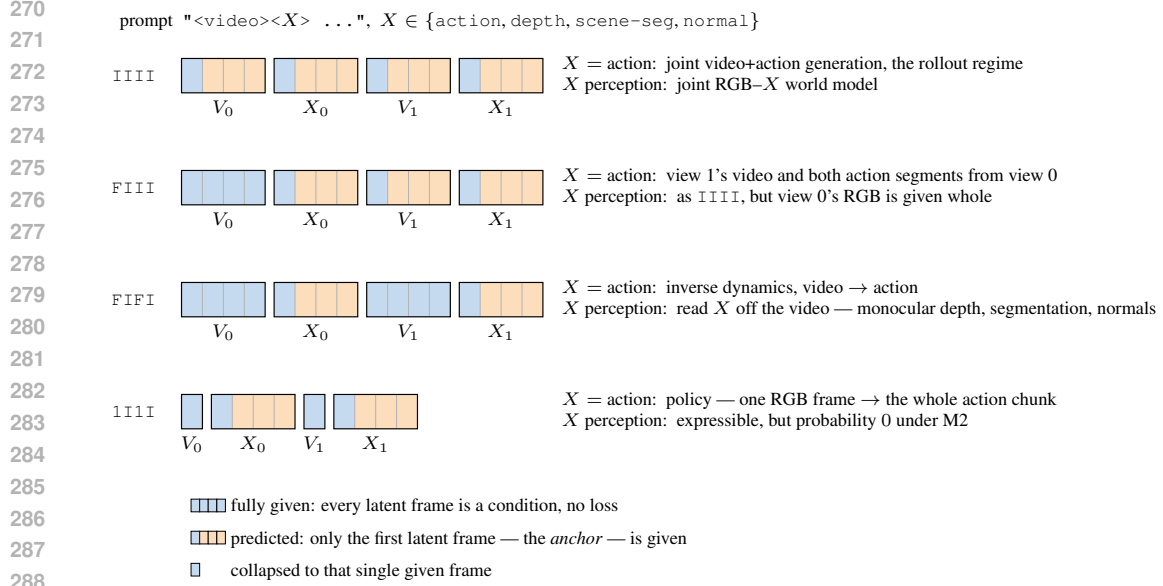


Figure 1: Every conditioning plan the mixture draws, and what each one means in the two template families. Each segment is drawn as its strip of latent frames, because that is where the conditioning lives: a predicted segment still receives its *first* latent frame clean, as an anchor, and only the rest is denoised. The mask is a property of *positions*, not of contents, so one diagram per plan covers all four auxiliary streams — only the task it names changes, which is the content of the $(\Pi, \mathbb{G})$ factorisation. The last row also shows that segments need not share a length: under 1II1 each visual segment collapses to its single anchor frame while the other stream keeps full length.

The mixture draws four plans, and each is worth reading twice — once per template family — because that is where the factorisation earns its keep (figure 1). II II gives every segment only its anchor: with an action stream this is joint video-and-action generation, the regime the closed-loop rollout runs in; with an auxiliary stream it is a joint RGB- X world model. FIFI gives both RGB segments whole, and the same two templates become inverse dynamics (video $\rightarrow$ action) and monocular estimation of X . FIII sits between them, giving only view 0’s RGB so that view 1 must be produced as well. 1II1 collapses each RGB segment to its single anchor while the other stream keeps full length: one frame in, a whole action chunk out. Two families, four plans, eight tasks — and one loss. The probabilities are matched across the families and are given in section 4.4.

The point is that figure 1 needs four diagrams and not sixteen. A plan constrains *positions* in the packed sequence, and depth, segmentation and normal all take the identical branch, so the mask a plan produces does not depend on which stream occupies the slot — only the name of the resulting task does. *Perception in the sense of Gabeur et al. (2026) and Wang et al. (2026) is therefore not the presence of an auxiliary stream; it is the pair $(\Pi = \{\text{video}, X\}, FIFI)$.* The same Π under II II is a multimodal world model, not an estimator of X . And a template that substitutes depth for RGB entirely, $[D_0 \mid A_0 \mid D_1 \mid A_1]$, contains no RGB, so nothing can be read off it: expressible, but a depth-space world model, and reporting it as perception would be a category error. Making that distinction statable rather than implicit in a data loader is the main reason we insist on the $(\Pi, \mathbb{G})$ factorisation.

4.4 THE TRAINING MIXTURE

A step draws a template, then a plan, then builds the sample (figure 1). Our headline arm uses

$$\Pi \sim [\text{video+action:0.4, video+depth:0.2, video+segmentation:0.2, video+normal:0.2}]. \quad (11)$$

The plan distribution is conditioned on whether Π contains an action. For action-bearing templates we transcribe the upstream distribution decision-for-decision, so the baseline stays reproducible; the

realised marginals are `IIII` 72.9%, `IIII` (policy) 9%, `FIII` 4.05%, `FIFI` 4.05% and video-only 10% (appendix B).

For perception templates the plan distribution is a free parameter, and we argue an important one. The obvious choice, $M0 = (\text{IIII } 0.10, \text{FIII } 0, \text{FIFI } 0.90)$, gives the RGB in full almost always, so the auxiliary modality must be *read off* the video rather than hallucinated; it is what makes `video+depth` an honest depth estimator. We nonetheless default to $M2 = (0.90, 0.05, 0.05)$, matching the action-bearing split, for two reasons. (a) *Template identity should not predict conditioning level*. Under $M0$, `<depth>` in the prompt implies “the RGB will be given”, so the model can read the conditioning pattern off the tag — a shortcut that transfers nowhere. (b) *Deployment runs IIII*. The closed-loop rollout generates `video+action` from first frames only, so an auxiliary stream trained 90% under `FIFI` learns a discriminative RGB→depth map in a regime the deployed model never enters. Section 5.3 measures the price of this choice, which is real.

4.5 CONSISTENCY BY CONSTRUCTION

Prompt-switched multitask training has a characteristic failure mode: a sample silently becomes a different task than the mixture specification claims, and every instrument stays green, because equation 6 is a plain MSE against whatever target was actually built. Three invariants, each corresponding to a failure we hit.

Prompt and layout are decided together. Upstream decides action dropout inside the forward pass, by testing whether the action tensor happens to be all zeros. We move that decision into the dataset, where the prompt is built; otherwise the text promises `<action>` for a sequence containing none. The same rule makes a dataset that cannot serve a requested template fail at *startup* rather than quietly serve a shorter one.

Ground truth comes from the same window and views as the RGB. View directories are enumerated by an unsorted glob and two of four are sampled per sample, so re-deriving the mapping afterwards yields a different draw. The sampled frame indices and view directories are first-class fields of the sample, which makes it structurally true — not merely checked — that a depth segment and an RGB segment of one view are the same scene, camera and instants.

Annotation gaps are startup errors, not training-time degradations. When 70% of episodes lacked segmentation targets, a nominal 60/20/20 action/depth/seg mixture silently realised as 60/20/6/14, the last 14% being video-only samples carrying *neither* action nor perception supervision. A second incident painted 5.2% of all pixels with a reserved colour that must never occur in training data (appendix D). Both are now startup gates. We report them not as engineering trivia: in a mixture-of-tasks study, a silent 14% shift in the realised mixture is larger than any effect we are trying to measure.

5 EXPERIMENTS

5.1 SETUP

Data. A self-generated RLBench (James et al., 2020) tree at 512^2 with Colosseum-style domain randomisation (Pumacay et al., 2024) over camera pose, table colour and texture, background texture and light colour: 16 tasks, 4 views, per-frame metric depth and handle masks. The split is by *variation*: variation-0 (788 episodes) is trained on and all other variations are held out as unseen task configurations. Windows are 41 frames at stride 3 (6.0 s of motion at 6.7 Hz); two of four views are sampled per sample.

Training. All arms warm-start from the released Action Images checkpoint (step 125,750) and are otherwise identical — same seed, step count, learning rate, data tree and prompt style — so any difference is attributable to the task menu. Full-parameter fine-tuning of the 5B DiT with ZeRO-2 and CPU-offloaded Adam, bf16, batch size 1 per device on 2 GPUs, learning rate 5×10^{-7} , 1000 warmup steps, constant schedule. Arms: **arm0** `video+action@1.0` (control), **arm4** 0.6/0.4 action/depth, **arm1** 0.6/0.2/0.2 action/depth/segmentation, and the headline **arm6** of equation 11.

Metrics. AbsRel for depth, macro-mIoU over the roles present for segmentation, mean cosine for normals, and for action the median position and geodesic rotation error of the decoded trajectory.

Table 2: Four modalities from one arm6 checkpoint (step 1500), switched by prompt tag alone. Each row is a separate generation from the same weights on the same held-in episode (open_drawer, view 0). $n=1$ episode; Table 3 gives the cross-episode spread.

Prompt	Metric	IIII	FIII	FIFI
<video><action>	$r_{\text{peak}} \uparrow$ (of 255)	252.9	252.8	252.1
<video><depth>	AbsRel $\downarrow$	0.087	0.059	0.652
<video><scene-seg>	macro-mIoU $\uparrow$	0.386	0.354	0.343
<video><normal>	mean cos $\uparrow$	0.896	0.919	0.937
<video>	PSNR (dB) $\uparrow$	16.8	30.7 [†]	n/a [‡]

[†] segment is fully given; the value is the VAE round-trip, not a prediction.

[‡] video is its own anchor, so FIFI would violate equation 10.

Table 3: Depth across four held-in episodes at arm6 step 4500. FIFI — the “easy” discriminative mode, where both RGB views are given in full — receives 5% of perception probability mass under M2, and it is the mode that fails, on some episodes only. IIII, the deployment regime, is stable everywhere.

Episode	IIII AbsRel $\downarrow$	FIII AbsRel $\downarrow$	FIFI AbsRel $\downarrow$	given-RGB PSNR
open_drawer	0.160	0.069	1.335	30.7
push_buttons	0.065	0.103	0.041	38.2
stack_wine	0.116	0.106	0.123	31.5
turn_tap	0.059	0.029	0.035	34.6
mean	0.100	0.077	0.384	—

We also report r_{peak} , the per-frame maximum of the red channel: a liveness signal the loss does not provide, since a low r_{peak} means the model has stopped drawing a decodable blob at all. Closed-loop is task success under receding-horizon replanning.

5.2 ONE CHECKPOINT, FOUR PROMPTS

Table 2 is the central versatility result: 1500 steps ($\approx 1\%$ of the original training budget) suffice for one checkpoint to answer four different questions about the same scene, selected by four different prompts, with the same weights and the same head. Depth is decodable to 8.7% AbsRel *while the RGB it must be consistent with is itself being predicted*; normals reach 0.90–0.94 cosine against a VAE ceiling of 0.99; nine-way dense role segmentation reaches 0.39 macro-mIoU; and the action stream keeps $r_{\text{peak}} \approx 253$ with zero weak frames, i.e. the policy is intact. A second arm confirms the pattern is not mixture-specific: arm1 at step 2000 gives, under IIII over both views, depth AbsRel 0.088/0.114 and segmentation macro-mIoU 0.426/0.566.

5.3 THE CONDITIONING AXIS IS NOT FREE

Table 3 isolates the cost of the M2 plan distribution. Under IIII — 90% of perception draws, and the regime the policy is deployed in — depth is stable across episodes (0.059–0.160) and across training (0.087 at step 1500, 0.086 at 3500, 0.085 at 4000). Under FIFI, which receives 5% of the mass, the same checkpoint gives AbsRel 1.34 on open_drawer and 0.04 on push_buttons: the under-trained mode has become episode-dependent, and on the one episode we track through training it degrades monotonically (0.652 $\rightarrow$ 1.167 $\rightarrow$ 1.335), while FIII oscillates instead (0.059, 1.303, 0.069), which reads as instability rather than forgetting (appendix C). Segmentation moves the other way over the same interval (0.386 $\rightarrow$ 0.449 under IIII).

This is the central trade of the framework. The plan distribution is not a hyperparameter that only affects convergence speed; it decides *which questions the finished checkpoint can still answer*. M0 buys a strong depth estimator and a weak world model, M2 the opposite. A mixture that wants both must pay for both, and on 788 episodes we could not.

Table 4: Action quality of the four-stream checkpoint (arm6, step 4500) under every action-bearing mode it was trained on, decoded to 6-DoF poses (open_drawer, view 0, $n=1$ episode). More given video yields better action, which is the ordering the conditioning axis predicts.

Mode	pos. err. med. (cm) ↓	pos. err. mean (cm) ↓	rot. err. med. (°) ↓	frac. $>90^\circ$ ↓	r_{peak} ↑
FIFI	1.9	2.3	18.7	0.049	250.8
FIII	4.5	7.4	19.8	0.049	252.4
IIII	5.4	6.4	20.9	0.049	254.1
policy	14.8	14.5	26.9	0.268	251.0

Table 5: Closed-loop success on 5 RLBench tasks, $n=20$ scenes each ($n=100$ total), paired by scene seed, McNemar per task. “official” is the shared warm-start checkpoint, so each column is the net effect of fine-tuning *relative to its own initialisation*, not an external baseline. The fine-tuned arm is in-domain and the official checkpoint is zero-shot on this rig; this is a home-advantage comparison, not a model ranking.

Task	official (init)	arm4 @5k	arm4 @10k	McNemar b/c	p
close_box	80%	70%	60%	3/5	0.727
close_drawer	5%	45%	80%	8/0	0.0078
push_buttons	5%	40%	20%	8/1	0.039
meat_off_grill	10%	0%	15%	0/2	0.500
open_drawer	0%	0%	0%	0/0	1.000
total	20%	31%	35%	19/8	0.052

5.4 THE POLICY SURVIVES, AND KEEPS ITS MODES

Table 4 shows the four-stream checkpoint still produces usable trajectories under every action-bearing mode. The ordering is exactly the information ordering — FIFI sees both videos in full and solves an inverse-dynamics problem, IIII must invent the future it acts in, and policy sees a single frame — which is a sanity check that the conditioning axis does what it claims. That a checkpoint whose training mixture is 60% non-action still recovers 1.9 cm inverse dynamics is our strongest evidence that the shared head is not being torn apart by the auxiliary streams.

5.5 CLOSED-LOOP SUCCESS, AND WHAT THE LOSS DOES NOT TELL YOU

Table 5 reports closed-loop success for the depth co-training arm. Two tasks improve significantly over the initialisation (+40 and +35 pp, both $p < 0.05$), two move the other way without significance, and the pooled effect sits exactly on the threshold ($p = 0.052$, 95% CI [+1, +21] pp). We report it that way rather than rounding: per-task effects of opposite sign cancel, and reporting only “31% vs 20%” would hide both real effects. The action-only control (arm0) is flat from step 750 to 2500 (20–30%, overlapping confidence intervals).

Loss does not order checkpoints. At step 7500, arm4’s success falls to 6.7% ($n=60$) from 28.8% at 2500 and 36.7% at 10000 (Fisher $p = 1.0 \times 10^{-4}$ against 10000), and the collapse is present on every task, not one. Over the same interval the training loss is at its *lowest* of the whole run (mean 0.0486 over steps 7000–7500 and 0.0457 over 7500–8000, versus 0.0487 immediately before), with no outlier steps. A transient exists that costs $5\times$ the success rate and then recovers on its own, and the diffusion objective is completely blind to it. Practically: checkpoint selection by loss is invalid here, dense checkpointing plus rollout evaluation is mandatory, and any single-step comparison between two arms is untrustworthy.

6 LIMITATIONS

Scale. 788 training episodes, 16 simulated tasks, a single seed. Zhuang et al. (2025) report accuracy loss on *every* task once a multitask vision model’s task set outgrows its data, and we sit three orders

of magnitude below their data budget — which is why our headline arm caps the auxiliary set at three streams. We cannot separate “four streams is the right number” from “four streams is what 788 episodes can carry”.

Evaluation breadth. The modality tables are $n=1$ episode for three of the four streams and $n=4$ for depth, on *held-in* variation-0 scenes; the 478 held-out episodes have not been swept. Table 5 is $n=100$ paired scenes but measures the action/depth arm, not the four-stream arm, whose training had reached step 4500 of a planned 10000.

The tag is not isolated from the anchor. Every mode we evaluate gives the target segment its first latent frame as a clean anchor, which is itself an instance of the requested modality. The prompt tag and the anchor therefore always agree, and we have not established how much of the modality selection each one carries. Isolating that requires training-time exposure to tag-only conditioning, which the current mixture does not contain.

One gate is not measured. The action stream’s codec floor (≈ 4 mm 3D) is known, but its VAE round-trip is not: unlike the perception streams, the action canvas is rendered inside the forward pass from the trajectory and both cameras rather than loaded from disk, so measuring it needs a harness that re-renders, round-trips and then re-runs the multi-view pose fusion, which the three perception gates do not share. We report that cell as unmeasured rather than inferring it from the RGB control.

Untrained cells, and simulation only. Equation 9 expresses three-modality templates and substitution templates such as $[D_0 \mid A_0 \mid D_1 \mid A_1]$; neither is trained here, and both are stated as reachable structure rather than as results. Depth and role masks are simulator ground truth, so extending to real data means pseudo-labels — and our own probing found monocular pseudo-depth on in-the-wild robot video to lack the metric anchor that equation 7 requires.

7 CONCLUSION

A world action model can co-generate action, depth, dense semantic roles and surface normals through a single shared output head, with the modality selected by a prompt tag and the task type — perception versus world model — selected by an orthogonal conditioning mask. The construction costs no new parameters and, for two-modality templates, no extra compute, because an auxiliary modality occupies the slot the action would have used. The $(\Pi, \mathbb{G})$ factorisation is what makes claims in this area falsifiable: it separates a depth-space world model from monocular depth estimation, and it exposes the conditioning mixture as a design decision that determines which questions a finished checkpoint can still answer. That point, together with the finding that diffusion loss does not order checkpoints by closed-loop success, is what we would most want a reader building in this direction to take away.

REPRODUCIBILITY STATEMENT

All arms are launched by a single script parameterised only by the template mixture; the data tree, seed, step count, learning rate, warm-start checkpoint and prompt style are held fixed across arms and stamped into the output path, so that two differently-configured runs cannot share a directory. The codec admission gate, the mixture probabilities, the exact command lines, and the artifact from which each reported number was extracted are given in the appendix and in the code release.

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A CODEC DETAILS

Depth. The RGB-cube path of equation 7 visits black, red, yellow, green, cyan, blue, magenta, white — eight vertices and seven unit edges, one coordinate flipping per step. Encoding clips to a sanity range of 0.05–10 m and writes an off-path sentinel (128, 0, 128) outside it. Decoding solves

$$(k^*, \alpha^*) = \arg \min_{k \in \{0, \dots, 6\}, \alpha \in [0, 1]} \left\| \gamma - ((1 - \alpha)\mathbf{v}_k + \alpha\mathbf{v}_{k+1}) \right\|^2, \quad \hat{y} = \frac{1}{7}(k^* + \alpha^*), \quad (12)$$

and inverts the transform as $\hat{d} = \lambda c (1 - (1 - \hat{y})^{1/(\lambda+1)})$. Any pixel whose residual in equation 12 exceeds a threshold is reported as invalid rather than returned as a plausible depth.

Surface normal. In a right-handed camera frame with pinhole focal lengths (f_x, f_y) in pixels,

$$\mathbf{n} \propto \left(-\frac{f_x}{z} \frac{\partial z}{\partial u}, -\frac{f_y}{z} \frac{\partial z}{\partial v}, 1 \right)^\top, \quad \text{stored as } \frac{1}{2}(\mathbf{n} + 1), \quad (13)$$

following the visualisation convention of Zhuang et al. (2025). The f/z factors are not optional: taking ∇z in pixel units, as an earlier version of our own code did, is wrong by $f_x/z \approx 200$ –750 on this data and flattens every surface systematically toward the camera — while producing entirely plausible-looking normal maps. Derivatives are central differences, one-sided at the border. We deliberately emit *no* invalid sentinel: the simulator renders sky at a constant far-clip depth that equation 7 already encodes as an ordinary surface, and a constant-depth region has zero gradient, hence maps to $(0, 0, 1)$, a camera-facing plane. The two streams must not disagree about what the sky is.

Segmentation. Nine roles with globally fixed colours: `background` (black), `target` (red), `goal` (green), `robot_arm` (blue), `gripper` (cyan), `distractor` (yellow), `tool` (magenta), `fixture` (purple), `unknown` (orange). Two constants are calibrated rather than chosen: the `role` $\rightarrow$ `colour` assignment is permuted so that the tightest pair in sRGB distance (98.3) lands on `unknown`, a class that must never occur in valid data, and the background gate $\tau = 40$ is set against that measured minimum rather than a nominal one. Encoding is a `handle` $\rightarrow$ `role` $\rightarrow$ `colour` lookup, so roles cannot overlap and there is no paint order to get wrong; decoding is nearest-colour classification restricted to the roles the episode can legally contain, and uses *no* connected-component pruning: the standard instance-decode post-processing deletes legitimate few-pixel targets outright (e.g. `sweep_to_dustpan`’s dirt, five handles of 4–8 px). A handle’s role is its instruction-independent base role, then promotion to `target` for handles the instruction refers to *and* whose base role is a manipulable object — a referred receptacle stays a `goal` and a referred implement stays a `tool`, since 9 of the 16 tasks name two or three objects in a single instruction.

B REALISED MIXTURE PROBABILITIES

For an action-bearing template draw (upstream values, transcribed decision-for-decision):

Outcome	Composition	Probability
video-only (action dropout)	0.10	10.0%
policy (single-frame visual)	0.90×0.10	9.0%
IIII	$0.90 \times 0.90 \times 0.90$	72.9%
FIII	$0.90 \times 0.90 \times 0.05$	4.05%
FIFI	$0.90 \times 0.90 \times 0.05$	4.05%

For perception templates the presets over (IIII, FIII, FIFI, single-frame anchor) are $M0 = (0.10, 0, 0.90, 0)$, $M1 = (0.45, 0.10, 0.45, 0)$ and $M2 = (0.90, 0.05, 0.05, 0)$; all experiments here use M2. The single-frame branch means something different for the two families: with an action present, collapsing every visual segment still leaves the action segments to predict, whereas in an all-visual template collapsing everything would violate equation 10, so only the anchor is collapsed.

C AUXILIARY STREAMS OVER TRAINING

Step	depth II II ↓	depth F I I I ↓	depth F I F I ↓	seg II II ↑	seg F I F I ↑
1500	0.087	0.059	0.652	0.386	0.343
3500	0.086	—	—	—	—
4000	0.085	1.303	1.167	0.449	0.592
4500	0.160	0.069	1.335	—	—

`arm6`, `open_drawer`, view 0. Depth under the deployment regime II II is flat, segmentation improves, and the low-probability FIFI depth mode degrades. The training loss over the same interval is smooth and predicts none of it.

D TWO SILENT-FAILURE INCIDENTS

Realised-mixture drift. `seg_targets.json` is generated offline, so every episode added by the scene generator arrives without it. An episode with no segmentation target resolves nothing, at which point the sample drops `segmentation` from its template and silently becomes a plain `video` one. Measured over the tree at the time, 70% of variation-0 episodes were affected (97% of the newly generated ones), turning a nominal 60/20/20 action/depth/seg mixture into roughly 60/20/6/14 — where the final 14% carried neither action nor perception supervision. The loss curve was normal throughout, the construction-time self-test passed (it probes episodes until one works), and the run’s own logs reported whatever mixture was on the command line. The loader now refuses to start above a 10% miss rate.

An unmapped handle in the segmentation target. The simulator renders “no object” (the sky above the room walls) as `-1`, which the mask writer converts to `0xFFFFFFFF` and then truncates to `0xFFFF` on the cast to `uint16`. That value reached the `handle → role` lookup unmapped and was painted with the reserved unknown colour: 5.2% of all pixels over 11,200 frames, up to 52.9% in a single view. Nothing flagged it, because equation 6 is a plain MSE against whatever target was built, and the one test that did check pointed at a data root whose symlink had gone stale. A startup probe over a spread of episodes now checks it and cannot be skipped.

E NUMBER PROVENANCE

Every number in this paper is extracted from a stored artifact rather than transcribed from a log; the mapping from table cell to artifact path is listed in `paper/README.md` in the code release. The modality grid comes from a single script that loads the checkpoint once and sweeps every (template, mode) pair, recomputing segment spans from the same rule the inference path uses and asserting the total against the array actually returned, so that a slicing disagreement cannot silently move a metric onto the wrong pixels.